A2 · Unit 4 · WPH14 | Further Mechanics, Fields & Particles

Nuclear & Particle Physics

What is everything made of? A century ago the answer was "atoms"; then it was "electrons, protons and neutrons"; now it is quarks and leptons. Each step came from firing particles at other particles, harder and harder, and reading the tracks that came out. This page follows that story: alpha scattering, electron beams, accelerators, detectors, and the conservation laws that decide which particle reactions can happen.

SPEC 111 Nucleon number and proton number

WHAT THIS SPEC POINT ASKS

Understand what nucleon number (mass number) and proton number (atomic number) mean.

Nuclide notation

A nucleus of element X is written ${}^{A}_{Z}X$, for example ${}^{238}_{92}U$.

Symbol	Name	Meaning
A (top)	**nucleon number** (mass number)	total number of **protons + neutrons**
Z (bottom)	**proton number** (atomic number)	number of **protons**; fixes which element it is
N = A − Z	neutron number	number of neutrons

- Uranium-238: 92 protons, 238 − 92 = **146 neutrons**. A neutral atom also has 92 electrons.
- **Isotopes** have the same Z but different A (same number of protons, different numbers of neutrons), e.g. ${}^{1}_{1}H$, ${}^{2}_{1}H$, ${}^{3}_{1}H$.
- An **ion** has gained or lost electrons; its nucleus, and so A and Z, are unchanged.
- Particle symbols in this notation: proton ${}^{1}_{1}p$, neutron ${}^{1}_{0}n$, electron ${}^{0}_{-1}e$, positron ${}^{0}_{+1}e$, alpha ${}^{4}_{2}He$.

Balancing nuclear equations

Both totals must match on each side: the **top numbers** (nucleon number) and the **bottom numbers** (charge).

WORKED EXAMPLE

balancing equations

Alpha decay of radon-222: $^{222}_{86}Rn \rightarrow {}^{218}_{84}Po + {}^{4}_{2}He$. Top: 222 = 218 + 4 ✓; bottom: 86 = 84 + 2 ✓.

An alpha particle hitting beryllium: $^{4}_{2}He + {}^{9}_{4}Be \rightarrow {}^{12}_{6}C + {}^{1}_{0}n$. Top: 4 + 9 = 12 + 1 ✓; bottom: 2 + 4 = 6 + 0 ✓.

MARK-SCHEME WORDING

Often asked as: MCQ counting protons and neutrons from the symbol (e.g. A = 24, Z = 10 → 14 neutrons); completing a nuclear equation.

COMMON MISTAKE

Swapping proton and neutron numbers: every MCQ has the reversed pair as a distractor.

Reading "nucleon number" as "neutron number". Nucleons are protons **and** neutrons.

SPEC 112 Alpha scattering and the nuclear model of the atom

WHAT THIS SPEC POINT ASKS

Understand how large-angle alpha-particle scattering gave evidence for a nuclear atom, and how ideas about atomic structure have changed.

The experiment (Geiger and Marsden, explained by Rutherford, 1909–1911)

- A narrow beam of **alpha particles** was aimed at a very **thin gold foil** in an **evacuated** chamber.
- A detector (originally a zinc sulfide screen viewed through a microscope, giving tiny flashes) was moved round the foil to count alpha particles at each scattering angle.

Observation	Conclusion
Most alpha particles went **straight through**, undeflected	the atom is **mostly empty space**
Some were deflected through **small angles**	there is a **concentration of (positive) charge** that repels the alphas

Observation	Conclusion
A **very few** (about 1 in 8000) were deflected by **more than 90°**, some straight back	the positive charge and **most of the mass** are concentrated in a **tiny region**: the **nucleus**

- The count rate falls very steeply as the scattering angle increases.

Why the plum-pudding model failed

Thomson's model (1904) spread positive charge and mass **evenly** through the atom, with electrons dotted in it. That diffuse charge could only produce **small** deflections. Only a small, dense, charged nucleus can produce a force strong enough to turn an alpha particle back.

Details of the set-up

- **Vacuum:** so alpha particles are not absorbed or scattered by **air molecules** (their range in air is only a few cm).
- **Thin foil:** so each alpha is scattered by **one** nucleus at most (no multiple deflections) and is not absorbed.
- **Gold:** can be hammered extremely thin; its large nuclear charge (79e) gives big deflections.
- **Alpha particles**, not beta or gamma: beta particles are much less massive and are deflected by the atomic **electrons**; gamma rays are **uncharged**, so no electrostatic deflection. Both are also too penetrating.

How the model of the atom changed

Date	Model	Key idea
~1800	Dalton	atoms are tiny indivisible spheres
1897–1904	Thomson	discovered the electron; plum-pudding model
1911	Rutherford	small, dense, positive nucleus; electrons outside
1913	Bohr	electrons in fixed energy levels (explains line spectra)
1932	Chadwick	discovered the neutron, inside the nucleus
1960s–70s	high-energy electron scattering	protons and neutrons are made of quarks

WORKED EXAMPLE

An alpha particle with kinetic energy 5.0 MeV is fired directly at a gold nucleus (Z = 79). Calculate its distance of closest approach. What does this tell you about the size of the gold nucleus?

Show solution

At the closest point the alpha stops for an instant, so all its kinetic energy has become electric potential energy: $E_k = \dfrac{Q_1 Q_2}{4\pi\varepsilon_0 r}$.

$E_k = 5.0 \times 1.60 \times 10^{-13} = 8.0 \times 10^{-13}$ J, $Q_1 = 79e$, $Q_2 = 2e$:

$$r = \frac{8.99 \times 10^9 \times (79 \times 1.60 \times 10^{-19}) \times (2 \times 1.60 \times 10^{-19})}{8.0 \times 10^{-13}} = 4.5 \times 10^{-14} \text{ m}$$

The alpha never touched the nucleus, so the radius of a gold nucleus must be **less than** about 4.5×10^{-14} m. (It is actually about 7×10^{-15} m.)

MARK-SCHEME WORDING

The six-mark answer pairs each observation with its conclusion:

- "most went straight through (undeviated), so the atom is mostly **empty space**"
- "a few deflected through small angles, so there is a **concentration of charge**"
- "very few deflected by **more than 90°**, so most of the **mass** is concentrated in a very **small** region (nucleus)"
- plum-pudding "predicted only small deflections"; the new model has "a very small nucleus containing almost all the mass and charged".

"Thin foil so alpha particles are less likely to undergo **multiple deflections**" or "less likely to be absorbed".

Often asked as: 6-mark description and significance; why vacuum, thin foil, gold; why not beta or gamma; MCQ on which statement was **not** a conclusion.

COMMON MISTAKE

Writing "deflected through angles **less than** 90°" for the large-angle evidence. It is **more than 90°**.

Listing observations without the conclusion each supports.

Saying "the nucleus contains protons and neutrons" was a conclusion. Neutrons were not discovered until 1932.

"Reflected off the nucleus": say **repelled** by the positive nucleus.

⚠ Beyond the spec · not part of the Edexcel IAL specification

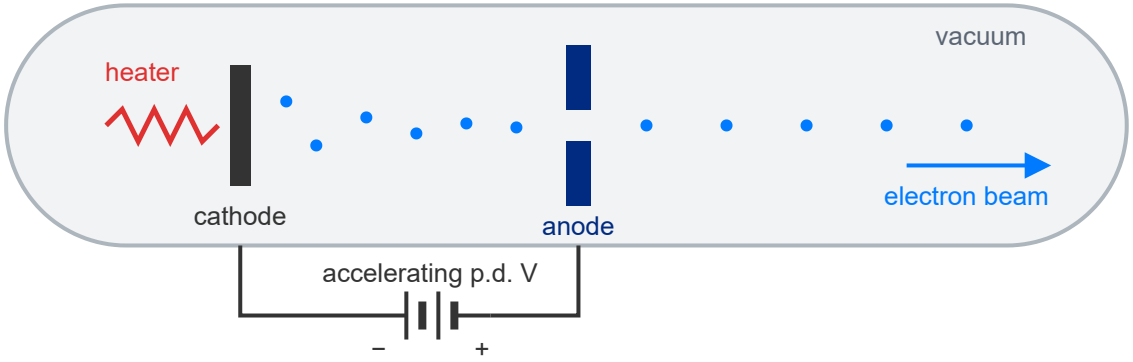
Examined in past papers even though the spec doesn't list it.

- **Closest approach** gives an upper limit for the nuclear radius: set the alpha's kinetic energy equal to the electrical potential energy, Ek = (2e)(79e)/4πε₀r (see 98; WPH14 Jun 2022, Oct 2023).
- **Momentum in scattering:** an alpha bouncing off a light nucleus (or proton) is a 2-D collision problem (WPH14 Jun 2023).

SPEC 113 Thermionic emission and accelerating electrons

WHAT THIS SPEC POINT ASKS

Electrons are released by thermionic emission and can be accelerated by electric and magnetic fields.



An electron gun. The anode is positive relative to the cathode, so the electric field between them accelerates the emitted electrons. Beyond the anode there is no field, so they travel at constant velocity.

Thermionic emission

- A metal **filament** (or cathode) is **heated** by passing a current through it.
- Some of the free (conduction) electrons gain enough energy to **escape from the metal surface**. This is **thermionic emission**.
- It must happen in a **vacuum**, so the electrons do not collide with gas molecules.
- A hotter filament (bigger heater current or heater p.d.) emits **more** electrons per second, giving a **larger beam current**.

Accelerating them: the electron gun

- A positive **anode** near the cathode creates an **electric field** that accelerates the electrons. Many pass through a hole in the anode as a beam.
- The work done on each electron by the accelerating p.d. V becomes kinetic energy:

Accelerating electrons

$$eV = \tfrac{1}{2}mv^2$$

e: electron charge, 1.60×10^{-19} C V: accelerating p.d. / V m: electron mass / kg
v: final speed / m s⁻¹

- e = 1.60 × 10⁻¹⁹ C, V = accelerating p.d. (V), m = 9.11 × 10⁻³¹ kg, v = final speed (m s⁻¹).
- A larger anode p.d. gives a **faster** beam; it does not change how many electrons are emitted.
- A **magnetic** field cannot change the electrons' speed: its force is perpendicular to their velocity, so it only changes their **direction** (steering and focusing beams, or bending them into circles).

WORKED EXAMPLE

speed from an electron gun

Electrons are accelerated through 2.0 kV.

Ek = eV = 1.60 × 10⁻¹⁹ × 2.0 × 10³ = 3.2 × 10⁻¹⁶ J

v = √(2Ek/m) = √(2 × 3.2 × 10⁻¹⁶ ÷ 9.11 × 10⁻³¹) = **2.7 × 10⁷ m s⁻¹**

That is 9% of c, so the non-relativistic formula is still fine.

WORKED EXAMPLE

Electrons are emitted from a hot cathode and accelerated through a p.d. of 2.0 kV. Calculate their kinetic energy in joules and their speed.

Show solution

$E_k = eV = 1.60 \times 10^{-19} \times 2.0 \times 10^{3} = 3.2 \times 10^{-16}$ J (that is 2.0 keV)

$v = \sqrt{\frac{2E_k}{m}} = \sqrt{\frac{2 \times 3.2 \times 10^{-16}}{9.11 \times 10^{-31}}} = 2.7 \times 10^{7}\ \text{m s}^{-1}$, about 9% of the speed of light. At much higher p.d.s this simple equation stops working because the speed cannot exceed c.

MARK-SCHEME WORDING

- Two marks: "the (metal) **filament is heated** (by the current)" and "the electrons are released by **thermionic emission**" (or "electrons gain enough energy to leave the surface").
- DO NOT accept "thermal effect".
- "Use of Ek = QV" and "use of Ek = ½mv²"; some questions require at least 3 s.f.

Often asked as: name and describe the emission; speed of electrons from a given p.d.; MCQ on how to increase the beam current.

COMMON MISTAKE

Not naming thermionic emission, or not saying what is heated.

Saying the accelerating p.d. increases the number of electrons emitted.

Thinking a magnetic field can speed electrons up.

SPEC 114 Particle accelerators and detectors

WHAT THIS SPEC POINT ASKS

Understand the roles of electric and magnetic fields in linear accelerators, cyclotrons and particle detectors (ionisation and deflection only).

The linear accelerator (linac)

- A line of hollow metal **drift tubes**, connected alternately to the two terminals of a high-frequency **alternating p.d.** of **constant frequency**, all inside a vacuum.
- Particles are accelerated by the **electric field in the gaps** between tubes.
- Inside a tube there is **no electric field** (the tube shields it), so the particle **drifts at constant speed**.
- The p.d. **reverses while the particle is inside a tube**, so when it reaches the next gap the field there is again in the accelerating direction.
- The time spent in each tube is the same (half a period of the a.c.), but the particle gets faster, so **each tube must be longer** than the last (length = v × T/2).

- For electrons close to the speed of light the speed hardly increases any more, so the tubes stop getting longer (the energy still increases).

WORKED EXAMPLE

drift-tube length

A proton moving at 2.0×10^6 m s^{-1} enters a tube; the a.c. frequency is 25 MHz.

Time in the tube = T/2 = $1 \div (2 \times 25 \times 10^6) = 2.0 \times 10^{-8}$ s

Length = vt = $2.0 \times 10^6 \times 2.0 \times 10^{-8}$ = **0.040 m**

The cyclotron

- Two hollow D-shaped electrodes (**dees**) in a vacuum, with a uniform **magnetic field perpendicular** to the dees and an **alternating p.d.** across the gap between them.
- **Electric field** (in the gap only): **accelerates** the particles each time they cross the gap.
- **Magnetic field**: exerts a force **perpendicular to the velocity**, which provides the **centripetal force**, so inside each dee the particle moves in a **semicircle** (no change of speed).
- After each crossing the particle is faster, so the radius (r = mv/BQ) is bigger: the path is an outward **spiral**. It leaves at the edge.
- Time for a semicircle: T/2 = πm/BQ. This **does not depend on speed or radius**, so the p.d. can switch at a **constant frequency** f = BQ/2πm, reversing while the particle is inside a dee.
- At speeds approaching c the particle takes longer per revolution (a relativistic effect), falls out of step with the p.d., and the cyclotron stops working.

WORKED EXAMPLE

a proton cyclotron

B = 0.80 T, dee radius 0.40 m, proton.

Period T = $2\pi m/BQ = 2\pi \times 1.67 \times 10^{-27} \div (0.80 \times 1.60 \times 10^{-19}) = 8.2 \times 10^{-8}$ s, so f = 1/T = **12 MHz**

Exit speed: v = $BQr/m = 0.80 \times 1.60 \times 10^{-19} \times 0.40 \div 1.67 \times 10^{-27} = 3.1 \times 10^7$ m s^{-1}

Ek = $\frac{1}{2}mv^2 = 7.8 \times 10^{-13}$ J = **4.9 MeV**

With 20 kV across the gap, each crossing adds 20 keV, so about 245 crossings, or 123 turns (two crossings per turn).

Detectors

- **Ionisation:** a fast charged particle knocks electrons out of atoms along its path. The ions are made visible as a **track** (droplets in a cloud chamber, bubbles in a bubble chamber) or detected as electrical signals (wire and

spark chambers, semiconductor detectors).

- **Neutral particles do not ionise**, so they leave **no track**. Their presence is inferred (e.g. from a gap, or from missing momentum).
- **Deflection:** a magnetic field across the detector bends the tracks. The **direction** of bending gives the **sign of the charge**; the **radius** gives the **momentum** (r = p/BQ).
- Heavily ionising particles (alphas, slow particles) leave thick, short tracks; fast electrons leave thin tracks.

Why the cyclotron frequency doesn't depend on speed

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 Inside a dee the particle moves in a semicircle of radius r at speed v. The time to go half way round is:

$$t_{1/2} = \frac{\pi r}{v}$$

2 The radius is $r = mv/BQ$ (derived in the next section). Substitute:

$$t_{1/2} = \frac{\pi}{v} \cdot \frac{mv}{BQ} = \frac{\pi m}{BQ}$$

The speed cancels. A faster particle goes round a bigger circle, and the two effects exactly balance.

3 One cycle of the supply must last two half-turns, so the period is $2\pi m/BQ$ and the frequency is:

$$\boxed{f = \frac{BQ}{2\pi m}}$$

So a fixed-frequency supply stays in step with the particle on every turn, from the centre to the edge.

WORKED EXAMPLE

A cyclotron for protons has dees of radius 0.50 m in a magnetic field of 1.5 T. Calculate (a) the frequency of the alternating supply and (b) the maximum kinetic energy of the protons, in MeV.

Show solution

(a) $f = \frac{BQ}{2\pi m} = \frac{1.5 \times 1.60 \times 10^{-19}}{2\pi \times 1.67 \times 10^{-27}} = 2.3 \times 10^{7}$ Hz (23 MHz)

(b) The largest orbit has $r = 0.50$ m, so the maximum momentum is $p = BQr = 1.5 \times 1.60 \times 10^{-19} \times 0.50 = 1.2 \times 10^{-19}$ kg m s^{-1}.

$$E_k = \frac{p^2}{2m} = \frac{(1.2 \times 10^{-19})^2}{2 \times 1.67 \times 10^{-27}} = 4.3 \times 10^{-12} \text{ J} = \frac{4.31 \times 10^{-12}}{1.60 \times 10^{-13}} \text{ MeV} = 27 \text{ MeV}$$

MARK-SCHEME WORDING

Cyclotron (the five points):

- "an **electric field** (from the p.d.) **between the dees** accelerates the particles"
- "the magnetic field is **perpendicular** to the velocity, so the **force is perpendicular** to the velocity"
- "this provides the **centripetal force**, so the path in a dee is a semicircle"
- "the p.d. switches polarity **when the particle is inside a dee**, so it is accelerated each time it crosses the gap"
- "constant frequency, because T = 2πm/BQ is **independent of speed** and radius"

Linac: "accelerated by the **electric field in the gaps**", "no field inside the tubes: constant speed", "a.c. of constant frequency, polarity changes while the particle is inside a tube", "time in each tube constant, so tubes get **longer** as speed increases", "near c the speed is almost constant, so tubes stop getting longer".

Detectors: "only charged particles leave tracks", "tracks formed by ionisation", "radius decreasing means the particle is losing energy".

Often asked as: 6-mark explanation of a linac or cyclotron, or a comparison of the two; cyclotron period, frequency or exit speed; sketch the spiral path; identify neutral particles from gaps in tracks.

COMMON MISTAKE

Saying the magnetic field accelerates the particles (it only changes their direction).

Saying "the p.d. accelerates them" without mentioning an **electric field**, or not saying **where** the field is.

Not saying **when** the polarity changes.

Using one dee for a whole period: the time in one dee is **T/2**.

Describing the cyclotron without explaining why the frequency can be constant.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Deriving T = 2πm/BQ from r = p/BQ and v = 2πr/T (WPH14 Jan 2020).
- Counting gap crossings: energy per crossing = QV; two crossings per revolution (WPH14 Jan 2020, Jun 2025).

- **Synchrotron radiation:** charged particles moving in circles emit radiation and lose energy, so the radius shrinks (WPH14 Jun 2024 MCQ).

SPEC 115 Radius of a charged particle's path: r = p/BQ

WHAT THIS SPEC POINT ASKS

Derive and use r = p/BQ for a charged particle moving at right angles to a magnetic field.

Derivation

The magnetic force is perpendicular to the velocity, so it provides the centripetal force:

Charged particle in a magnetic field

FORMULA SHEET

$$r = \frac{p}{BQ}$$

r: radius of path / m p: momentum / kg m s^{-1} B: flux density / T Q: charge / C

Deriving r = p/BQ

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. A particle of charge Q moves at speed v at right angles to a uniform field B. The magnetic force on it is:

$$F = BQv$$

This force is always perpendicular to the velocity, so it provides the centripetal force and the particle moves in a circle.

2. Set the magnetic force equal to the centripetal force:

$$BQv = \frac{mv^2}{r}$$

3. Cancel one factor of v and rearrange for r:

$$r = \frac{mv}{BQ}$$

4 mv is the momentum p, so:

$$\boxed{r = \frac{p}{BQ}}$$

A larger momentum gives a larger, less curved track; a stronger field or a larger charge curls it up more tightly.

- r = radius (m), p = momentum (kg m s^{-1}), B = flux density (T), Q = charge (C).
- **Faster or heavier** particles (more momentum) make **bigger** circles. **Stronger fields** or **larger charges** make **smaller** circles.
- A particle losing energy (e.g. by ionising) slows down, so its radius **decreases**: a **spiral** track.
- The time for one orbit, T = 2πm/BQ, does not depend on the speed.

WORKED EXAMPLE

electron beam in a field

Electrons from a 2.0 kV gun (v = 2.65 × 10^{7} m s^{-1}, from 113) enter a 2.5 mT field at right angles.

p = mv = 9.11 × 10^{-31} × 2.65 × 10^{7} = 2.41 × 10^{-23} kg m s^{-1}

r = p/BQ = 2.41 × 10^{-23} ÷ (2.5 × 10^{-3} × 1.60 × 10^{-19}) = **0.060 m**

WORKED EXAMPLE

an alpha particle

A 5.0 MeV alpha particle (m = 6.64 × 10^{-27} kg, Q = 2e) moves at right angles to a 1.2 T field.

Ek = 5.0 × 10^{6} × 1.60 × 10^{-19} = 8.0 × 10^{-13} J; p = √(2mEk) = √(2 × 6.64 × 10^{-27} × 8.0 × 10^{-13}) = 1.03 × 10^{-19} kg m s^{-1}

r = 1.03 × 10^{-19} ÷ (1.2 × 3.20 × 10^{-19}) = **0.27 m**

WORKED EXAMPLE

A proton leaves a circular track of radius 0.25 m in a bubble chamber with a magnetic field of 1.2 T. Calculate its momentum, in SI units and in MeV/c, and its kinetic energy in MeV.

Show solution

$p = BQr = 1.2 \times 1.60 \times 10^{-19} \times 0.25 = 4.8 \times 10^{-20}\ \text{kg m s}^{-1}$

In MeV/c: $pc = 4.8 \times 10^{-20} \times 3.00 \times 10^{8} = 1.44 \times 10^{-11}\ \text{J} = 90\ \text{MeV}$, so $p = 90\ \text{MeV}/c$.

$E_k = \dfrac{p^2}{2m} = \dfrac{(4.8 \times 10^{-20})^2}{2 \times 1.67 \times 10^{-27}} = 6.9 \times 10^{-13}\ \text{J} = 4.3\ \text{MeV}$. Much less than its rest energy (938 MeV), so the non-relativistic formula is fine.

MARK-SCHEME WORDING

- Derivation: "equates F = mv²/r and F = BQv", "substitutes p = mv with suitable algebra to arrive at r = p/BQ".
- Track photographs: "determines the radius from measurements", "**applies the scale**".
- Momentum units: N s or kg m s⁻¹, or GeV/c.

Often asked as: derive r = p/BQ; find B or p from a radius; radius from a photograph with a scale; MCQ (four times the mass, twice the charge, same speed → radius doubles).

COMMON MISTAKE

Using the diameter as the radius.

Using the electron mass for a proton.

For an alpha particle, using Q = e instead of 2e.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Very fast electrons:** p = mv no longer works. Use **E = pc** (E in J) to get the momentum (WPH14 Jun 2021).
- **Mass spectrometer:** ions of the same charge and speed but different mass follow circles of different radius, so isotopes separate (WPH14 Oct 2022).

SPEC 116 Conservation laws and particle tracks

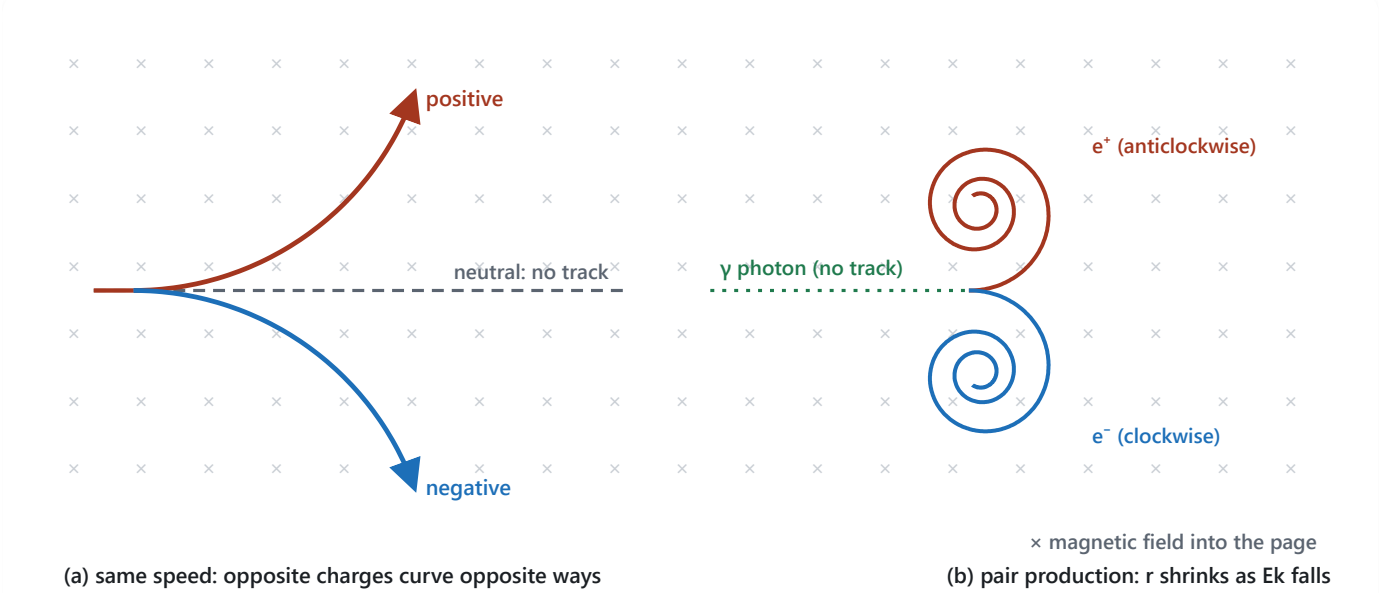
WHAT THIS SPEC POINT ASKS

Apply conservation of charge, energy and momentum to particle interactions, and interpret particle tracks.

What is conserved in every interaction

- **Charge:** total charge before = total charge after.
- **Energy** (mass–energy): total of rest energies + kinetic energies before = total after. Kinetic energy can become mass and mass can become kinetic energy ($\Delta E = c^2\Delta m$).
- **Momentum:** vector sum before = vector sum after. If a particle at rest decays into two, they fly apart in **opposite** directions. A momentum vector diagram must **close**.

Reading a track photograph



Field into the page. (a) At the same speed, positive and negative particles curve opposite ways (Fleming's left-hand rule); a neutral particle leaves no track and is not deflected. (b) A gamma photon (no track) makes an electron–positron pair: two tracks from one point, curving opposite ways, spiralling inwards as they lose energy.

What you see	What it tells you
Track curves	particle is charged; curvature direction + field direction (FLHR) → **sign** of charge
Large radius	large **momentum** (r = p/BQ)
Radius shrinks along the track (spiral)	particle is **losing energy** and slowing; it travels **towards** the tighter curl
Track bends more after passing through a plate	it lost energy in the plate, so it was travelling **into** the more curved part
Tracks appear from nowhere ("V" shape)	a **neutral** particle (no track) decayed there

What you see	What it tells you
Two tracks from one point, curving opposite ways	pair production, or decay into opposite charges
Track suddenly stops or kinks	the particle decayed (perhaps into neutral particles) or collided
Momentum before ≠ sum of visible momenta after	a **neutral** particle carried the missing momentum

WORKED EXAMPLE

field direction from a track

A positron moves to the right across the page and curves **upwards**. Which way is the field?

The positron is positive, so the second finger (current) points right. The thumb (force) points up the page (towards the centre of curvature). The first finger then points **into the page**: the field is into the page.

An electron in the same place would curve **downwards**.

MARK-SCHEME WORDING

- Particle deduction (6-mark style): "rest mass-energy of the incoming particles **plus their kinetic energy** = total after"; "some of the kinetic energy becomes **mass** ($\Delta E = c^2\Delta m$)"; "the momentum of the incoming particle equals the **vector sum** of the products (x-components), and the y-components **cancel**".
- Momentum diagram: "vectors drawn **end to end**", "arrows", "three lines form a **closed triangle**, so momentum is conserved".
- "Radius **decreasing**, so the particle is slowing / losing energy".

Often asked as: identify a missing (neutral) particle using charge and momentum; deduce field direction or charge sign from curvature; explain why a neutral particle leaves no track; draw a momentum triangle for a decay.

COMMON MISTAKE

Saying "the curvature decreases" when you mean the **radius** decreases (curvature increases).

Drawing a momentum triangle with the wrong particle's momentum in it.

Forgetting that neutral particles are invisible, so a gap in the picture can be a real particle.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

Using the gap at the start of a track, or the order of tracks, to decide which event happened first (WPH14 Oct 2020).

SPEC 117 Why high energies are needed to probe nucleons

WHAT THIS SPEC POINT ASKS

Understand why high energies are needed to investigate the structure of nucleons.

The resolution argument

- To see detail of a certain size, you need a probe whose **wavelength** is about that size or smaller (otherwise it just diffracts around the object).
- Moving particles have a **de Broglie wavelength**:

de Broglie wavelength

$$\lambda = \frac{h}{p}$$

λ: wavelength / m h: Planck constant, 6.63×10^{-34} J s p: momentum / kg m s^{-1}

- λ = wavelength (m), h = Planck constant = 6.63 × 10^{-34} J s, p = momentum (kg m s^{-1}).
- A nucleon is about **10^{-15} m** across. To resolve its structure, λ must be about 10^{-15} m or less, so the probe needs a **very large momentum**, and so a **very high energy**.
- **Electrons** are used as the probe: they are **leptons**, so they do not feel the strong force, and they are point-like, so they probe the charge inside cleanly.

WORKED EXAMPLE

energy needed

For λ = 1.0 × 10^{-15} m: p = h/λ = 6.63 × 10^{-34} ÷ 1.0 × 10^{-15} = **6.6 × 10^{-19} kg m s^{-1}**

These electrons are highly relativistic, so E ≈ pc = 6.63 × 10^{-19} × 3.00 × 10^{8} = 2.0 × 10^{-10} J ≈ **1.2 GeV**.

(Using ½mv² would give a speed far above c, which is impossible.)

A second, different reason: making new particles

- To **create** a massive particle, the collision must supply at least its rest energy, mc^2 (see 118).
- **Colliding beams** are better than a fixed target: two equal beams head-on have **zero total momentum**, so the products need not carry momentum or kinetic energy away, and **all** the energy is available to make new mass. With a fixed target, much of the energy stays as kinetic energy of the products.

MARK-SCHEME WORDING

- Structure: "high energy → **large momentum** → **small (de Broglie) wavelength**", "wavelength comparable to the size of the nucleon ($\approx 10^{-15}$ m)", "better resolution".
- Colliding beams: "no resultant momentum before the collision, so the products can have little momentum and kinetic energy, so all the energy is available to form the mass of new particles".

Often asked as: why high energy is needed to probe structure; MCQ where "to create new particles" is the distractor; why colliding beams are used; is the energy enough to make a particle of a given mass.

COMMON MISTAKE

Answering a "structure" question with $E = mc^2$. That explains making particles, not resolving detail.

Not linking all three: energy → momentum → wavelength → resolution.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Electron **diffraction** shows that electrons behave as waves: "an electron does not always behave as a particle" (WPH14 Jan 2023).
- **E = pc** for high-energy electrons (WPH14 Jun 2021).

SPEC 118 Creation and annihilation of matter and antimatter

WHAT THIS SPEC POINT ASKS

Use $\Delta E = c^2 \Delta m$ for the creation and annihilation of matter and antimatter particles.

Mass–energy

FORMULA SHEET

$$\Delta E = c^2 \Delta m$$

ΔE: energy / J Δm: mass / kg $c = 3.00 \times 10^8$ m s^{-1}

- ΔE = energy (J), Δm = mass (kg), c = 3.00 × 10⁸ m s⁻¹. Mass and energy are interchangeable.
- **Rest energy** of a particle = mc² (energy it has because of its mass alone).

Annihilation

- A particle meets its **antiparticle**: both disappear and their total energy (rest energy + kinetic energy) becomes **gamma photons**, usually **two**.
- If they were at rest, total momentum was zero, so the two photons move in **opposite directions** with **equal energies**. Each photon then carries the rest energy of **one** particle: E = mc².
- One photon alone cannot be produced, because it could not have zero momentum.

Pair production

- A high-energy **photon** turns into a **particle–antiparticle pair** (e.g. e^- and e^+).
- Minimum photon energy = **2mc²** (the rest energy of both particles). Any extra becomes their kinetic energy.
- It happens near a **nucleus**, which recoils to conserve momentum.

WORKED EXAMPLE

electron–positron annihilation

Rest energy of an electron: E = mc² = 9.11 × 10⁻³¹ × (3.00 × 10⁸)² = 8.20 × 10⁻¹⁴ J = 0.51 MeV.

Each of the two photons has E = 8.20 × 10⁻¹⁴ J, so λ = hc/E = 6.63 × 10⁻³⁴ × 3.00 × 10⁸ ÷ 8.20 × 10⁻¹⁴ = **2.4 × 10⁻¹² m** (gamma rays).

WORKED EXAMPLE

pair production

Minimum photon energy for an e^-e^+ pair = 2 × 0.511 MeV = 1.02 MeV.

A 1.50 MeV photon makes a pair: kinetic energy shared by the pair = 1.50 − 1.02 = **0.48 MeV**.

WORKED EXAMPLE

is there enough energy?

A collision makes 2.9 GeV available. Can it produce a proton–antiproton pair?

Rest energy needed = 2 × 1.67 × 10^{-27} × (3.00 × 10^{8})² = 3.0 × 10^{-10} J = 1.88 GeV.

1.88 GeV < 2.9 GeV, so **yes**; the remaining 1.0 GeV becomes kinetic energy.

WORKED EXAMPLE

(a) Calculate the minimum photon energy, in MeV and in J, needed to produce an electron–positron pair, and the wavelength of such a photon. (b) A proton and an antiproton, both at rest, annihilate into two identical photons. Calculate the energy of each photon in GeV.

Show solution

(a) $E = 2m_e c^2 = 2 \times 9.11 \times 10^{-31} \times (3.00 \times 10^8)^2 = 1.64 \times 10^{-13}\ \text{J} = 1.02\ \text{MeV}.$

$\lambda = \dfrac{hc}{E} = \dfrac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{1.64 \times 10^{-13}} = 1.2 \times 10^{-12}\ \text{m}$, a gamma ray.

(b) The total rest energy is $2 \times 938\ \text{MeV} = 1876\ \text{MeV}$, shared between two photons, so each has $938\ \text{MeV} = 0.938\ \text{GeV}.$

MARK-SCHEME WORDING

- "Use of ΔE = c²Δm", "conversion factor for eV", then a **comparison with a conclusion**: "1.88 GeV < 2.9 GeV so yes".
- Kinetic energy of products = energy available − rest energy of the products.
- Compare energies, not speeds.

Often asked as: mass of a particle in kg or GeV/c²; photon energy or frequency from annihilation or pair production; energy left over as kinetic energy; is a process possible.

COMMON MISTAKE

Not squaring c, or rearranging m = E/c² wrongly.

Forgetting the "M" (10^6) or "G" (10^9).

Forgetting to subtract the rest energy of the products, or to add the initial kinetic energy of the colliding particles.

Giving each annihilation photon the rest energy of **both** particles.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Finding the photon **frequency** from E = hf in pair production (WPH14 Jun 2021).
- Comparing a calculated particle mass with a table value as a percentage difference (WPH14 Jun 2025).

SPEC 119 Energy and mass units: MeV, GeV, MeV/c², GeV/c²

WHAT THIS SPEC POINT ASKS

Use MeV and GeV for energy and MeV/c², GeV/c² for mass, and convert between these and SI units.

The electronvolt

- **1 eV** is the energy gained by an electron accelerated through a p.d. of 1 V: 1 eV = 1.60 × 10^{-19} J.
- 1 keV = 10^{3} eV; **1 MeV = 10^{6} eV = 1.60 × 10^{-13} J**; **1 GeV = 10^{9} eV = 1.60 × 10^{-10} J**.
- eV → J: **multiply** by 1.60 × 10^{-19}. J → eV: **divide**.

Mass in MeV/c² and GeV/c²

- From m = E/c², a mass can be quoted as an energy divided by c².
- 1 MeV/c² = 1.60 × 10^{-13} J ÷ (3.00 × 10^{8})² = **1.78 × 10^{-30} kg**. 1 GeV/c² = 1.78 × 10^{-27} kg.
- **Momentum** in GeV/c: because GeV/c² × c = GeV/c, "GeV/c" is a unit of momentum.

Particle	Mass in kg	Mass in MeV/c²
electron	9.11 × 10^{-31}	0.511
proton	1.67 × 10^{-27}	938 (≈ 0.94 GeV/c²)
neutron	1.67 × 10^{-27}	940

WORKED EXAMPLE

conversions both ways

Proton, kg → MeV/c²: E = mc² = 1.67 × 10^{-27} × (3.00 × 10^{8})² = 1.503 × 10^{-10} J; ÷ 1.60 × 10^{-13} = **939 MeV**, so m ≈ 939 MeV/c² (the accepted 938 MeV/c² comes from more precise constants).

Top quark, GeV/c² → kg: 173 GeV = 173 × 10^{9} × 1.60 × 10^{-19} = 2.77 × 10^{-8} J; m = E/c² = 2.77 × 10^{-8} ÷ (3.00 × 10^{8})² = **3.08 × 10^{-25} kg**.

MARK-SCHEME WORDING

- "Conversion factor for eV" is its own mark.
- "GeV/c has the same unit as momentum, because GeV/c² × c = GeV/c".

Often asked as: MCQ mass in kg from MeV/c²; identifying GeV s^{-1} as a unit of power; converting a particle's mass for a comparison.

COMMON MISTAKE

Neglecting the M or G prefix.

Multiplying by c² when you should divide (E = mc² gives energy from mass; mass = E/c²).

SPEC 120 Relativistic increase in particle lifetime

WHAT THIS SPEC POINT ASKS

Understand when the relativistic increase in a particle's lifetime matters (no relativistic equations needed).

The muon puzzle

- **Muons** are made by cosmic rays hitting the upper atmosphere, about 10 km up. At rest their mean lifetime is about **2.2 μs**.
- Even at 0.98c, a muon would travel only s = vt = 0.98 × 3.00 × 10^{8} × 2.2 × 10^{-6} ≈ **650 m** in that time.
- Yet large numbers of muons **reach the ground**.
- Explanation: muons are moving **close to the speed of light**. For an **observer on Earth**, the muon's **lifetime is longer** (time dilation), so it travels much further before decaying.

When it matters

- Relativistic effects are tiny at everyday speeds. They become **significant** only at speeds that are a large fraction of c (roughly above about 0.1c, and large near c).
- In accelerators, unstable particles moving near c live longer in the **laboratory frame**, so they travel further and leave longer tracks before decaying.
- The same effects explain why ½mv² fails for very fast particles, and why a cyclotron falls out of step at high speeds.

WORKED EXAMPLE

classical distance

A pion with lifetime 2.6×10^{-8} s moves at 0.95c. Classically: $s = 0.95 \times 3.00 \times 10^{8} \times 2.6 \times 10^{-8}$ = **7.4 m**. If it is detected 50 m away, its lifetime in the lab frame must be longer than 2.6×10^{-8} s: a relativistic effect.

MARK-SCHEME WORDING

- "Muons travelling **close to the speed of light**" → "**relativistic** effect **increases** the particle's lifetime (as measured by the **observer**)" → "so it travels further than expected".
- "Time **dilation**" (do not accept "dilution").

Often asked as: 2-mark explanation of why a particle is detected far from where it was made; s = vt to show the classical distance is too short; MCQ (lifetime increases, rate of decay decreases).

COMMON MISTAKE

Mentioning time dilation but not saying what happens to the **lifetime**.

Saying the particle's own lifetime changes: it is the lifetime measured by the **observer** that is longer.

SPEC 121 The standard quark-lepton model

WHAT THIS SPEC POINT ASKS

Know that particles can be classified as baryons (three quarks), mesons (quark + antiquark), leptons (fundamental) and photons, and that the model's symmetry predicted the top quark.

Fundamental particles

- **Quarks** come in six flavours. They are never found on their own, only inside **hadrons**. They have **fractional** charges.
- **Leptons** are **fundamental**: they are not made of anything smaller. They do **not** feel the strong force. The electron, muon, tau and their three neutrinos are leptons.
- The **photon** is the particle (quantum) of electromagnetic radiation and carries the electromagnetic force. It has no charge and no mass, and is neither a hadron nor a lepton.

Hadrons (made of quarks; feel the strong force)

- **Baryons** = **three quarks** (antibaryons = three antiquarks). Examples: proton **uud** (charge +⅔ + ⅔ − ⅓ = +1), neutron **udd** (+⅔ − ⅓ − ⅓ = 0).
- **Mesons** = a **quark and an antiquark**. Examples: $\pi^+ = u\bar{d}$ (+⅔ + ⅓ = +1), $\pi^- = d\bar{u}$ (−⅓ − ⅔ = −1), $\pi^0 = u\bar{u}$ or $d\bar{d}$.
- Every hadron has a **whole-number** charge.

Quantum numbers

Particle	Symbol	Charge / e	Baryon number B	Lepton number L
up, charm, top quarks	u, c, t	+⅔	+⅓	0
down, strange, bottom quarks	d, s, b	−⅓	+⅓	0
antiquarks	$\bar{u}$, $\bar{d}$ …	opposite: −⅔ or +⅓	−⅓	0
proton	p (uud)	+1	+1	0
neutron	n (udd)	0	+1	0
antiproton	$\bar{p}$ ($\overline{uud}$)	−1	−1	0
pions	π^+, π^0, π^-	+1, 0, −1	0	0
electron, muon, tau	e^-, μ^-, τ^-	−1	0	**+1**
neutrinos	ν_e, ν_μ, ν_τ	0	0	**+1**
positron, antimuon	e^+, μ^+	+1	0	−1
antineutrinos	$\bar{\nu}_e$, $\bar{\nu}_\mu$	0	0	−1
photon	γ	0	0	0

- Baryon number: each quark +⅓, so any baryon (3 quarks) has B = +1; mesons (⅓ − ⅓) have B = 0.
- Lepton number: every lepton **+1** (including the negatively charged electron), every antilepton **−1**.

Generations and the top quark

Generation	Quarks	Leptons
1st	up (u), down (d)	electron (e^-), electron neutrino (ν_e)
2nd	charm (c), strange (s)	muon (μ^-), muon neutrino (ν_μ)
3rd	top (t), bottom (b)	tau (τ^-), tau neutrino (ν_τ)

The model groups the particles into three generations, each with **two quarks and two leptons**. After the bottom quark was found (1977) and the tau lepton was known, the **symmetry** of the model demanded a sixth quark to pair with the bottom quark. The **top quark** was predicted, and was found in 1995 at Fermilab.

MARK-SCHEME WORDING

- Baryon: "**3 quarks** (or 3 antiquarks)". Meson: "a **quark and an antiquark**" (ACCEPT $q\bar{q}$; DO NOT ACCEPT "quarks and antiquarks").
- Lepton: "fundamental / cannot be broken down", "L = ±1 and B = 0".
- Top quark: "a quark to match each lepton", "**by symmetry** of the standard model".
- "Mesons and baryons are **hadrons**."

Often asked as: definitions; build a meson or baryon from given quarks with the right charge; quark make-up of antiproton or pion; why the top quark was predicted; MCQ classification.

COMMON MISTAKE

Confusing a **muon** (a lepton) with a **meson**.

Writing "a baryon is made of quarks" without the number **three**.

Saying a neutron is "a proton plus an electron".

Giving a quark baryon number 0 (it is ⅓).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Charm, strange, top and bottom quarks and their charges (WPH14 Jun 2023, Jan 2023).
- **Strangeness:** the s quark has strangeness −1 ($\bar{s}$ has +1); questions give it when needed, e.g. building a strange meson $s\bar{u}$ with charge −1 (WPH14 Jan 2024).
- The muon is a second-generation lepton, about 207 times the electron's mass (WPH14 Jan 2022).

SPEC 122 Particles and antiparticles

WHAT THIS SPEC POINT ASKS

Every particle has an antiparticle; use a particle's properties to deduce its antiparticle's, and the other way round.

The rule

An antiparticle has:

- the **same mass** (and so the same rest energy),
- the **opposite charge**,
- the **opposite baryon number** and **opposite lepton number** (and opposite quark flavours).

Particle	Antiparticle	What changes
electron e^- (Q −1, L +1)	positron e^+ (Q +1, L −1)	charge and L
proton p (uud, B +1)	antiproton $\bar{p}$ ($\overline{uud}$, Q −1, B −1)	every quark → its antiquark
neutron n (udd)	antineutron $\bar{n}$ ($\overline{udd}$)	still neutral, but B = −1, so a different particle
neutrino ν_e (L +1)	antineutrino $\bar{\nu}_e$ (L −1)	lepton number
π^+ ($u\bar{d}$)	π^- ($\bar{u}d$)	swap quark ↔ antiquark
photon γ	photon γ	it is its own antiparticle

- Naming: usually "anti-" plus the name and a bar over the symbol. The **positron** is the exception in name.
- To get an antiparticle's quark content, replace **each** quark with its antiquark.

WORKED EXAMPLE

deducing an antiparticle

The Σ^+ baryon has quarks uus, charge +1, B = +1, L = 0, mass 1189 MeV/c².

Anti-Σ^+: quarks $\overline{uus}$, charge $-\frac{2}{3} - \frac{2}{3} + \frac{1}{3} =$ **−1**, B = **−1**, L = 0, mass **1189 MeV/c²**.

MARK-SCHEME WORDING

- "**Same mass**, **equal and opposite charge**, **opposite lepton number**" (for a lepton) or "opposite baryon number" (for a baryon).
- "DO NOT ALLOW lepton number of an electron = −1 or charge of an electron = +1": the **electron has L = +1**.
- Pair creation: two Ω particles would break baryon conservation (0 → 2); an Ω and an anti-Ω conserve it (0 → 1 − 1 = 0).

Often asked as: state two properties of an antiparticle; quark make-up of antiparticles; MCQ comparing a particle and its antiparticle.

COMMON MISTAKE

Giving the electron lepton number −1 because it is negative. Lepton number has nothing to do with charge: e^- is **+1**, e^+ is **−1**.

Getting the positron or neutrino lepton numbers wrong: the most common error in a recent report.

Changing only some quarks when forming an antiparticle.

SPEC 123 Conservation of charge, baryon number and lepton number

WHAT THIS SPEC POINT ASKS

Use conservation of charge, baryon number and lepton number to decide whether a particle interaction is possible.

The rule

In **every** interaction or decay, these totals are the same before and after:

- **charge** Q
- **baryon number** B
- **lepton number** L

(Energy and momentum are also always conserved; see 116 and 118.) If any one of Q, B or L is not conserved, the interaction **cannot happen**.

Some questions give separate **electron** and **muon** lepton numbers. If so, each family's lepton number must be conserved separately. That is why π^+ decays to $\mu^+ + \nu_\mu$, not $\mu^+ + \nu_e$.

How to set out the answer

1. Write each particle's value **under** it, including the zeros.
2. Add each side.
3. State "total before = total after, so conserved" (or "≠, so not conserved") for **each** quantity, then the conclusion.

WORKED EXAMPLE

beta-minus decay

$n \rightarrow p + e^{-} + \bar{\nu}_e$

Charge: $0 \rightarrow (+1) + (-1) + 0 = 0$ ✓

Baryon number: $1 \rightarrow 1 + 0 + 0 = 1$ ✓

Lepton number: $0 \rightarrow 0 + 1 + (-1) = 0$ ✓

All three are conserved, so the decay is possible. Without the antineutrino, L would be $0 \rightarrow 1$: **not** possible. This is why the antineutrino must exist.

WORKED EXAMPLE

an impossible reaction

$p + p \rightarrow p + p + \bar{p}$?

Charge: $(+1) + (+1) = 2 \rightarrow (+1) + (+1) + (-1) = 1$ ✗

Baryon number: $1 + 1 = 2 \rightarrow 1 + 1 + (-1) = 1$ ✗

Not possible. The real reaction makes a **pair**: $p + p \rightarrow p + p + p + \bar{p}$ (charge $2 \rightarrow 2$, B $2 \rightarrow 2$ ✓).

MARK-SCHEME WORDING

- "Lepton number: $0 + 0 \rightarrow 0 + 0 - 1 + 1$ (0 on each side, so conserved)": values for **every** particle, then the comparison.
- Marks come in pairs: "show the charge for all particles", then "total charge before = total charge after, so charge is conserved" (dependent on the first).
- Some questions want only two of the three laws ("MAX 2").

Often asked as: show that an interaction is (or is not) possible; identify which law it breaks; MCQ.

COMMON MISTAKE

"Baryon number is conserved because it is 2 before and 2 after" with no values for each particle: zero marks.

Writing "+1 + 1 + 0" but never showing the totals are equal.

Putting charge values in the lepton-number line.

Giving the neutron baryon number 0.

SPEC 124 Writing and interpreting particle equations

WHAT THIS SPEC POINT ASKS

Write and interpret particle equations using the relevant particle symbols.

Equations you should know

Process	Equation	Quark-level change
β^- decay	$n \rightarrow p + e^- + \bar{\nu}_e$	$d \rightarrow u + e^- + \bar{\nu}_e$
β^+ decay (in a nucleus)	$p \rightarrow n + e^+ + \nu_e$	$u \rightarrow d + e^+ + \nu_e$
electron capture	$p + e^- \rightarrow n + \nu_e$	$u + e^- \rightarrow d + \nu_e$
pair production	$\gamma \rightarrow e^- + e^+$	
annihilation	$e^- + e^+ \rightarrow \gamma + \gamma$	
pion decay	$\pi^+ \rightarrow \mu^+ + \nu_\mu$ and $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$	
alpha decay	${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$	

- In β^- decay a **lepton** (e^-) is made, so an **anti**lepton ($\bar{\nu}_e$) must be made too, keeping L = 0. In β^+ decay an antilepton (e^+) is balanced by a lepton (ν_e).
- Check every equation for charge, baryon number and lepton number (123), and for nuclear equations also the nucleon and proton numbers (111).

WORKED EXAMPLE

identify the unknown particle

$p + \bar{\nu}_e \rightarrow n + X$

Charge: $+1 + 0 = 0 + Q_X \rightarrow Q_X =$ **+1**

Baryon number: $1 + 0 = 1 + B_X \rightarrow B_X =$ **0**

Lepton number: 0 + (−1) = 0 + $L_X \rightarrow L_X$ = **−1**

A positive antilepton with no baryon number: X is a **positron, e^+**.

WORKED EXAMPLE

A negative pion collides with a stationary proton: $\pi^- + \mathrm{p} \rightarrow \mathrm{n} + \mathrm{X}$. Use conservation laws to identify X. Then explain why the reaction $\mathrm{p} \rightarrow \mathrm{n} + \pi^+$ cannot happen for a free proton.

Show solution

Charge: $-1 + 1 = 0 + Q_X$, so $Q_X = 0$. Baryon number: $0 + 1 = 1 + B_X$, so $B_X = 0$. Lepton number: $0 = 0 + L_X$, so $L_X = 0$.

X is uncharged with $B = 0$ and $L = 0$: a neutral meson such as the π^0 (or a photon).

For $\mathrm{p} \rightarrow \mathrm{n} + \pi^+$: charge (+1 = 0 + 1), baryon number (1 = 1 + 0) and lepton number (0 = 0) are all conserved. But the products have a total rest energy of $940 + 140 = 1080$ MeV, more than the proton's 938 MeV. A particle at rest cannot decay into heavier products, so energy conservation forbids it.

MARK-SCHEME WORDING

- Both particle **type** and particle/**anti**particle must be right: "there should be an **antineutrino** and not a neutrino".
- π^+ decay products: "$\mu^+ + \nu_\mu$"; the correct charged product is the **antimuon**.

Often asked as: write the equation for a decay given in words; complete an equation; identify X from conservation laws; MCQ on which equation is possible.

COMMON MISTAKE

Neutrino where an antineutrino is needed (or the reverse).

Charge not balanced, or the wrong sign on the muon.

Forgetting the gamma photons are two in annihilation at rest.

EXAM TIP

Do the lepton-number check last: it is the one that tells you whether you need a neutrino or an antineutrino.

Quick checks

QUICK CHECK

In the alpha-scattering experiment, what was the evidence that the nucleus contains most of the atom's mass?

- Most alpha particles passed straight through the foil.
- A few alpha particles were scattered back through more than 90°.
- The alpha particles were repelled, not attracted.
- The foil had to be very thin.

Only something much more massive than the alpha particle could turn it right round. Passing straight through shows the atom is mostly empty space; being repelled shows the nucleus is positive.

QUICK CHECK

A proton and an alpha particle move at the same speed at right angles to the same uniform magnetic field. What is the ratio (radius of alpha's path) / (radius of proton's path)?

- 1/2
- 1
- 2
- 4

$r = mv/BQ$. The alpha has 4 times the mass but twice the charge, so its radius is $4/2 = 2$ times as big.

QUICK CHECK

Why do the drift tubes in a linear accelerator get longer along its length?

- To give a larger accelerating p.d. at each gap
- The particle speeds up, but must spend the same time (half a cycle) inside each tube
- Because the particle's mass increases, so it needs more room
- To reduce the magnetic field needed to steer the beam

The supply frequency is fixed, so the particle must arrive at each gap exactly half a cycle after the last one. Faster particles cover more distance in that time, so each tube must be longer.

QUICK CHECK

Which of these is a meson?

- u u d
- u $\bar{d}$
- $\bar{u}$ $\bar{u}$ $\bar{d}$
- u d

A meson is a quark plus an antiquark: u $\bar{d}$ is the π^+. uud is a proton (baryon), $\bar{u}$ $\bar{u}$ $\bar{d}$ an antiproton (antibaryon), and u d is not allowed (charge +⅓).

QUICK CHECK

Which property of the electron antineutrino $\bar{\nu}_e$ is correct?

- Charge −1, lepton number +1
- Charge 0, lepton number +1
- Charge 0, lepton number −1
- Charge +1, lepton number −1

An antiparticle has the opposite charge and lepton number of its particle. The neutrino has charge 0 and $L = +1$, so the antineutrino has charge 0 and $L = -1$.

QUICK CHECK

Which reaction is impossible because baryon number is not conserved?

- $\pi^+ \rightarrow \mu^+ + \nu_\mu$
- $p + p \rightarrow p + p + \pi^0$
- $p \rightarrow \pi^+ + \pi^0$
- $e^- + e^+ \rightarrow \gamma + \gamma$

The proton has $B = +1$ but pions are mesons with $B = 0$, so B would go from 1 to 0. Charge is conserved (+1 = +1 + 0), but one failure is enough.